

ML Challenge 2026: Business Entity Resolution Solution Template

Team Name: Grenuke

Team Members: Ameya Borkar, Aarush Bakshi, Sachi Dhoka

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Contents

- | | |
|------------------------------------|--------------------------------|
| 1. Executive Summary | 5. Results & Error Analysis |
| 2. Methodology | 6. Conclusion |
| 3. Candidate Generation (Blocking) | Appendix A. Code Artefacts |
| 4. Matching Model | Appendix B. Additional Results |

1. Executive Summary

Our submission, which we call Composite B, scores **0.990879 macro F0.5 on the public leaderboard**. On our own **local validation** set, a held-out quarter of the labelled US and India data, the same pipeline scores **0.9913**. That set has no French records, because France has no labels.

The pipeline makes three passes. **Blocking** looks up candidate records for each Source-1 (S1) business one country at a time, and a gradient-boosted classifier cuts them down to **3.70 per S1** while keeping **99.1% of the true pairs**. More **gradient-boosted models** then score every candidate pair. On the pairs they are least sure about, fine-tuned multilingual **cross-encoders** help: e5, bge, and **Qwen2.5-1.5B and 7B** with LoRA. Finally, for each S1 we pick the set of records with the **highest expected F0.5**. There is no global threshold.

France was the hard part. It appears only in the test set, so we had nothing to train or validate on there. We **self-trained on our own French decisions**, with guards and cross-fitting so the model would not simply confirm its own mistakes. We also had a **Qwen2.5-7B model re-read the predictions** the pipeline was already certain about, and that second reading removed a group of **French decoys** that string features could not separate.

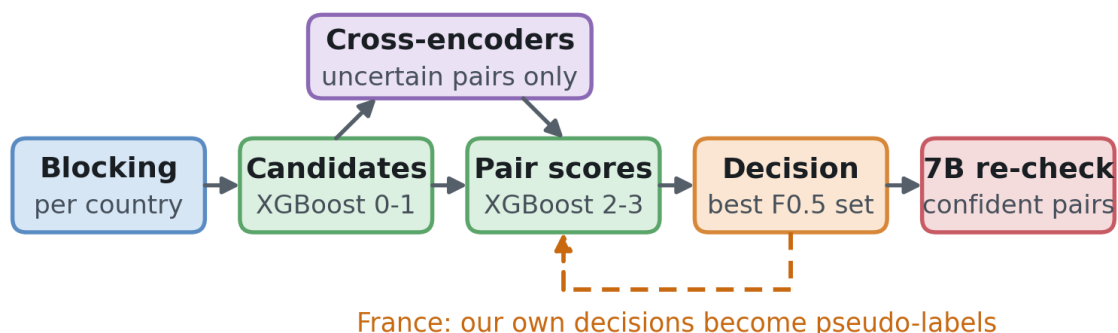


Figure 1. The pipeline, from records (left) to matches (right). The dashed line is the French self-training loop (Section 2.2).

2. Methodology

2.1 Problem Analysis

Each S1 business has between **0 and 11 copies** in the Source-2 and Source-3 files, 3.46 on average, and about **5.6% have none** at all. The metric is F0.5 averaged over S1, so **a wrong merge costs twice as much as a missed copy**. An S1 with no copies also scores zero as soon as we predict anything for it. That pushed the whole design towards **precision**.

The copies are **noisy**: case and accent changes, **legal forms** dropped or rewritten (about 23% of French copies), reordered or appended words, **acronyms**, typos, shortened **street types** ("Rue" to "R."), changed **house numbers** and **empty addresses**. Most of our blocking keys and features target one of these edits directly.

The test set also contains **decoys**. It has about **23% more records per S1** than the training set but the same number of true matches, so the extra records are **look-alikes**: a different business word at the same address, a nudged house number, or the same name somewhere else. Between **46% and 54% of S1 share their name** with another S1, so the **address** usually settles the match. We therefore let candidates **compete within each S1**, and each record can belong to **at most one S1**.

France is harder still: its names are **generic** ("lille club sarl"; each French decoy we removed shares its name with a median of **43 other S1**), **11%** of French S1 share their exact address with another, and **32%** of addresses carry a department or region name. We noticed early how much this mattered. Our first uploads scored **0.976 to 0.980** on the leaderboard while our local validation said **0.984 to 0.989**, and the only difference was France. Working backwards put **France at around 0.93**, and most of what we did afterwards was aimed at it.

2.2 Solution Strategy

Approach Type: Hybrid. **Blocking** generates candidates, three stages of **gradient-boosted classifiers** score the pairs, **cross-encoders** read the uncertain pairs, a **per-S1 decision** picks the final set, **rules** clean up known patterns, and an **LLM re-checks** the confident predictions (Figure 1).

Our strategy follows directly from the problem analysis. **Figure 2** maps each difficulty to what we did about it and what it gained, and the rest of this section explains the choices.

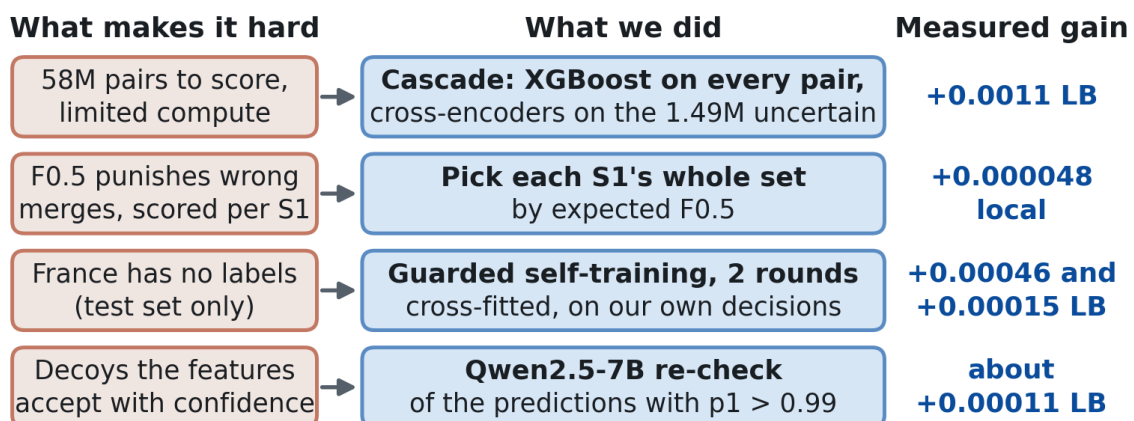


Figure 2. The solution strategy: each difficulty, what we did about it, and the gain we measured.

Two principles run through the whole design. The first is to **spend effort where the uncertainty is**. XGBoost is cheap enough to score all 58M retrieved pairs, the cross-encoders are affordable on the **1.49M pairs** where that score is uncertain, and the 7B model is used where a second opinion pays: in the mix, and as a reader of the predictions we were most sure about. The second principle is to **decide the way the metric scores**, per S1 and with precision first, so the last step chooses a set of records for each S1 rather than applying a threshold to pairs.

Core Innovation: three ideas carried most of the gain.

Self-training for a country without labels. We **pseudo-label French pairs** from our best model's final decisions. A pair it kept with a calibrated probability of **at least 0.9** becomes a positive, one it rejected at **0.05 or below** becomes a negative, and everything in between stays unlabelled. Two **guards** stop the model from reinforcing its own errors: where our rules decided a population, the rules win, and empty-address records keep their first-round labels. The labels are also **cross-fitted by S1 group**, so a French pair is never scored by a model trained on its own label. The first round added **+0.00046** on the leaderboard, and the second round, together with the French decision layers, another **+0.00015**.

A decision that optimises the metric directly. For each S1, a small **dynamic programme** over the calibrated candidate probabilities finds the prediction set with the **highest expected F0.5**. On our local holdout this beat the best global threshold by **+0.000048** (95% interval 0.000007 to 0.000091).

A second opinion on predictions we were sure about. **94.5%** of our final predictions have a stage-1 probability above 0.99, so **no cross-encoder ever looked at them**. A LoRA-tuned **Qwen2.5-7B** reads them again, and we drop the ones it rejects with a **logit below -6**, a cut-off we fixed on labelled data first. It dropped **840 French predictions**. On labelled data, only **8%** of predictions rejected that strongly are real matches.

Why not a simpler design. Each simpler option lost when we measured it. Scoring every retrieved pair with the 7B was out of reach: at the **roughly 600 pairs per second** we measured on an H100, the 58M test pairs alone would take **about 27 GPU-hours**, before any training. Carrying the US/India model over to France without self-training left the gap open, since the first round of self-training alone added **+0.00046**. And a **tuned global threshold** lost to the set selection on the local holdout.

How we decided what to keep. For US and India we held out a **fixed 25% of the training S1** (549,699 S1) and never trained on it. A component stayed only if it gained on that holdout in a **paired bootstrap**, and ties went to the simpler option. Rules added late also had to gain on **both halves** of the holdout. France has no labels, so there we relied on label-free checks, on estimates corrected with the 7B model's judgements, and on **leaderboard submissions that changed France alone**.

3. Candidate Generation (Blocking)

Blocking keys used: All searches run **separately per country**, so IDF weights and nearest neighbours never mix countries. Records without a country are searched in every partition. Table 1 lists the **three views**. The **token view** does most of the work. The **names-only view** covers records whose address is empty or very short, and the **repairs** fix names before any matching.

Table 1. Blocking views.

view	how it matches	what it catches
Token view	IDF-weighted overlap on name + address, both ways: each S1's top 40 records and each record's top 8 S1. A pair is kept if it is in the S1's top 15 or the record's top 4	most copies. A (house number, street word) key that skips street types handles French addresses
Names-only view	character 4-grams; single-token names of 8 or more letters	empty or short addresses, garbled names, domains and handles
Repairs	domain names split into the country's words; OCR digits in names fixed; an Indic-to-Latin dictionary of 693 entries learned from training pairs	web-style names, OCR errors, transliterated names

Candidate pairs generated: Retrieval produces **58.4M pairs** on the test set (66.4M on train). A first gradient-boosted scorer (stages 0 and 1) keeps every pair with $p1 \geq 0.02$, plus the **two best S1 for each record**, and we add pairs found by matching acronyms. The candidate file has **6,410,308 pairs, 3.70 per test S1**.

How you ensured true matches were not lost: We **measured recall** on the local holdout at every change instead of assuming it. The candidate set contains **99.1% of the true holdout pairs**. Most of the 16.5k it misses (out of about 1.9M) are **empty-address copies** whose name was also changed. Two **tighter cuts**, keeping only the best S1 per record or raising the threshold to 0.05, lowered holdout F0.5 by 0.000106 and 0.000041, so we kept the looser cut.

4. Matching Model

Features used: All statistics are **fitted per country** and there is **no country feature**, so nothing in the model is tied to US or India. Table 2 groups the features and says why each group is there.

Table 2. Feature groups.

group	main features	why
Name	fuzzy ratios (token-sort, token-set, partial, Jaro-Winkler); token Jaccard and containment; IDF-weighted cosine; legal-form agreement; counts of unexplained tokens; learned look-alike word odds	a copy keeps the rare words of a name, while a decoy swaps one business word
Address and house number	house-number equality, nudges of 1 to 10, digit edits and suffixes; street overlap; city and state; whole-address similarity; empty and PO-box flags	when two businesses share a name, the address tells them apart
Context	rank in each blocking view; the gap to the S1's best candidate and to rival S1; name-collision counts; per-S1 aggregates; cluster support; the cross-encoder score	a pair is judged against its competitors, not on its own

Model type: The pair classifiers are **four XGBoost stages**. Each is trained **out-of-fold** over three S1 groups, so every stage learns from honest scores of the one before it.

- **Stage 0** removes about **80% of pairs** cheaply.

- **Stage 1** produces the probability **p1**, which sets the candidate cut and decides which pairs go to the cross-encoders.
- **Stage 2** adds the cross-encoder score, per-S1 and cluster features, and the **French pseudo-labelled rows**. Its output is **calibrated** with isotonic regression.
- **Stage 3** re-scores each pair against the other candidates of its S1.

The **cross-encoders** only see the **uncertain band**: pairs with $0.02 \leq p1 \leq 0.99$, which is **1.49M** on the test set. Each is trained as three out-of-fold models on the text "name ; address" of both records, cut to 96 tokens. We **average their z-scored logits into one stage-2 feature**, because given separate scores, stage 2 extrapolated on pairs where the models disagree, and they disagree about **four times as often in France**.

Table 3 lists the models. French self-trained versions of e5-large and bge are also in the mix, and e5-base (278M, MIT) was used only for the first French teacher. The **7B model is no more accurate** than a self-trained e5-large; both score **0.944**. It earns its place in two ways: counted twice in the US/India mix it adds **diversity**, and it gives an **independent reading** of confident pairs in the re-check.

Table 3. Cross-encoders. Band AUC is measured on labelled pairs from the local holdout; stage-1 p1 alone reaches 0.930 on the same pairs.

model	size	licence	training	band AUC
multilingual-e5-small / e5-large	118M / 560M	MIT	full fine-tune	0.939 (large)
bge-reranker-v2-m3	568M	Apache-2.0	full fine-tune	0.942
Qwen2.5-1.5B	1.5B	Apache-2.0	LoRA	0.938
Qwen2.5-7B	7.6B	Apache-2.0	LoRA (r 16, alpha 32), three H100s for about 2 hours	0.944

Threshold selection method: There is **no single threshold**. The calibrated stage-3 probabilities go into the **expected-F0.5 selection** from Section 2.2. Three settings were tuned on the local holdout: a **logit shift of +0.2**, a further **−0.3** for records that four or more S1 compete for, and **0.01** expected true matches outside the candidate set. Each record is then assigned to its **highest-scoring S1**.

Rule layers follow. Acronym joins and per-source caps apply everywhere, and France also gets an **exact-address copy rule** (99.99% precise on US and India), a **cross-commune drop**, a **look-alike word-swap drop**, and the same set selection run on France's own probabilities. The last step is the **7B re-check at −6**. That cut-off had the **largest gain on the local holdout (+0.000037)**, positive in both halves and in 99.7% of random subsets.

5. Results & Error Analysis

F_0.5 Score (macro): Table 4 compares the **official leaderboard score** with our **local validation**. The local number is higher because **France cannot be part of it**, and most of the gap comes from France. The **private leaderboard**, which decides the final ranking, had not been published when we wrote this.

Table 4. Macro F0.5.

evaluation	macro F0.5
public leaderboard, official (US, India and France)	0.990879
local validation: labelled holdout, US and India only (549,699 S1)	0.9913 (US 0.9911, India 0.9916); precision 99.9%, recall 97.5%

Figure 3 shows how the leaderboard score grew over our submissions. The largest steps were the **e5-large cross-encoders (+0.0011)** and the **blocking repairs together with stage 3 (+0.0008)**. **French self-training** added +0.00046, and the last step, the **7B model in the mix plus the re-check**, added +0.00018.

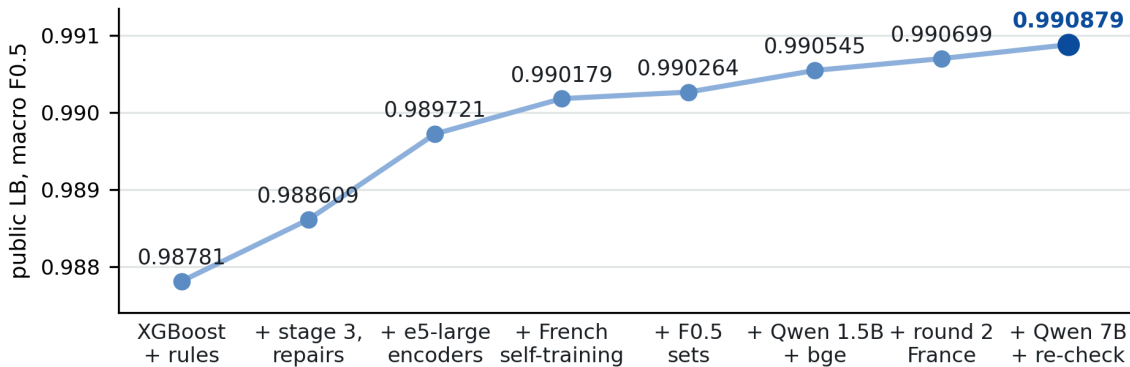


Figure 3. Public leaderboard score after each step. The last point is the submission in this package.

Common false positives (wrong merges): The most expensive errors were **generic-name decoys in France**: the same generic name and house number on a **different street**, such as "lille ecole sarl, 42 rue gutenber" against "lille ecole sarl, 42 q. du wault". On labelled data this pattern is a real match **0.5%** of the time when our model rejects it, and **99.7%** of the time when it accepts it. String features cannot tell the two apart, but **the 7B model can**, and removing its 840 French rejects was worth an estimated **+0.00011** of the final +0.00018. Smaller groups of errors are **business words swapped** at the same address ("antenne danse" and "antenne gaz"), very **short names** that differ by one letter ("osd" and "otd"), and **branches of a chain** that share a name.

Common false negatives (missed matches): Precision on the local holdout is **99.9%**, so most of the remaining loss is **recall**. **69%** of it comes from S1 where we found some copies but not all, and **20%** from S1 we missed completely. The missed copies are mostly records with an **empty address and a changed name**, which either never become candidates (16.5k) or get a score near zero (15.2k). **Heavy garbles**, such as the French "EHPAD" typed as "ehpvd", account for more. None of the recall rules we tried was more than **71% precise**, and under F0.5 an added pair must be right about **75%** of the time to help, so we did not add any.

6. Conclusion

On US and India, a **precision-first pipeline** with **calibrated per-S1 decisions** reached **0.9913** on our local validation. On the public leaderboard, which includes France, the submission scored **0.990879**. **Self-training** on our own guarded decisions got France most of the way without any labels. The **7B re-check** showed that a second, independent model can catch decoys that feature-based models accept with confidence.

Two lessons stay with us. Self-training helped for **two rounds**, and a third did not add to them. And on a country without labels, any estimate built from the model's own probabilities needs an **independent check**, because it shares the model's blind spots.

Appendix

A. Code Artefacts

`code/business_entity_resolution/` **regenerates both output files** from the raw train and test TSVs. Its `README.md` gives the exact commands, the run time of each step, and what each step should print.

`reproduce.sh` is the **entry point**; with `VARIANT=compositeB` (the default) it runs `src/box/compositeB.sh`. The team package is in `src/ber`, the model chain (XGBoost stages, cross-encoders, self-training labels, decision, rules, the France block) in `src/model_v1`, and the Qwen2.5-7B training, the 7B re-check and the per-country composition in `src/box`. The XGBoost stages need **24 or more CPU cores and 32 GB of RAM**, the cross-encoders ran on **one 80 GB H100**, and Qwen2.5-7B needs **three 80 GB GPUs** for about two hours. Exact versions are pinned in `requirements.txt`.

Seeds are fixed, and every artifact records the command and git commit that made it. GPU training is not bit-identical across machines (an earlier model rebuilt on other hardware moved from 0.991246 to 0.991261 on the holdout), so a rerun should land **within about 0.0001**. The `output/` folder holds the **exact bytes we submitted**.

B. Additional Results

Every local-validation gain quoted in this document was tested with a **paired bootstrap** on the labelled US/India holdout. Counting the 7B twice in the cross-encoder mix gained **+0.000066 on India** (P 0.998) and **+0.000026** on the US.

On a labelled 34% sample of the local holdout, **8.3%** of the confident predictions the 7B scored below -6 were real matches, against **92.6%** between -6 and -2 . It rejects **0.10% of French predictions** but only **0.004% to 0.007%** of US and India ones, at the same rate in each third of the French S1, including the third it never saw labels for. Two **leaderboard controls** agree: the round-3 French model without the French decision layers scored **0.990833**, and pushing the French 7B drops below -6 scored **0.990875**, a tie.

We also **measured and dropped** several ideas: **recall rules** (17% to 71% precise), a **learned blend** of all six cross-encoder scores as the decision score (0.001 lower F0.5), **mining extra drops** with a decision tree (it failed on the held-out half), **breaking ties** by copy count (-0.00057), **renormalising French probabilities** per record (up to -0.000137), and **other encoders** (Qwen3-4B was weaker than the 7B; mDeBERTa-v3 and gte-multilingual failed to train or run).